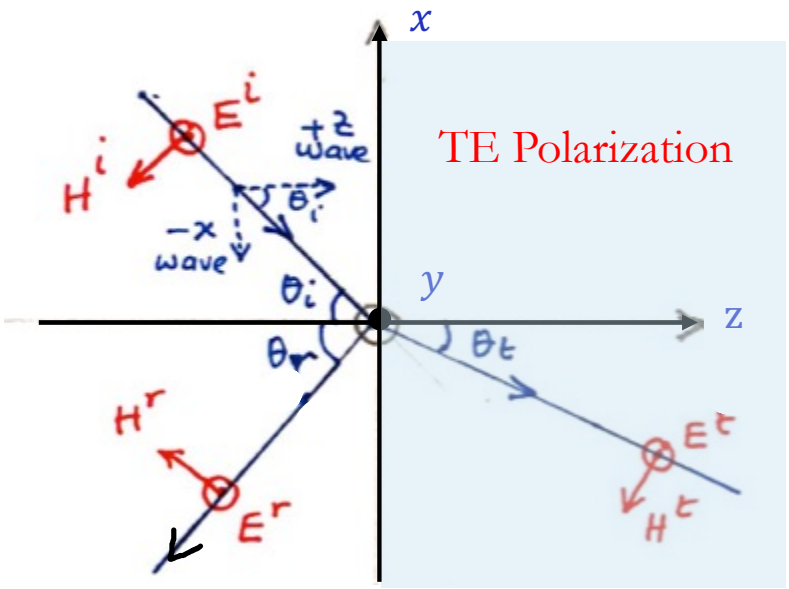
EM Waves Reflection and Transmission

The following subjects will be covered

- Single interface, Oblique Incidence.

Examples

A single-frequency (0.1 GHz) plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a dielectric constant of 2. The electric field of the incident wave has the following phasor form $\vec{E} = [\frac{1}{\sqrt{2}}\vec{a}_y]\, e^{-j\beta(z-\sqrt{2}x)/\sqrt{3}}$ V/m



From the $\vec{E}_i$ equation, the z incident wave takes the cosine. $\cos\theta_i = \frac{1}{\sqrt{3}}$

The x incident wave takes the sine. $\sin\theta_i = \frac{\sqrt{2}}{\sqrt{3}}$

$\theta_i = 54.7°$

From Snell's law, $\sin\theta_t = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}\sin\theta_i$ $\theta_t = 35.3°$

$\beta_o = 2.1$ rad/m $\beta_1 = 2.1\sqrt{2}$ rad/m

$$\sqrt{\varepsilon_1}\sin\theta_i = \sqrt{\varepsilon_2}\sin\theta_t$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

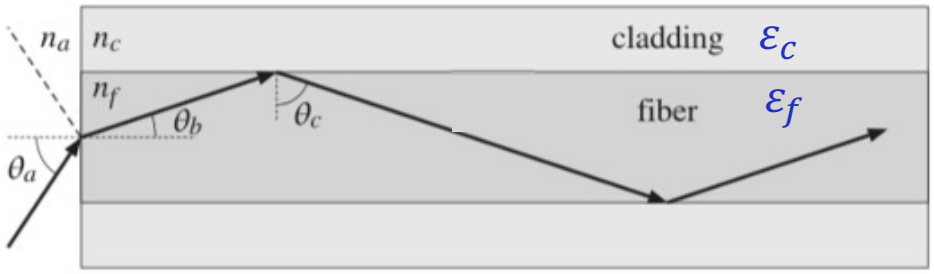
$$\vec{H}_r = \frac{(-0.333)}{(377)(\sqrt{2})}\, e^{j\beta_o(\sqrt{2}x+z)/\sqrt{3}}\, (\vec{a}_x - \sqrt{2}\vec{a}_z)/\sqrt{3} \text{ A/m}$$

$$\vec{E}_t = \frac{(0.667)}{(\sqrt{2})}\, e^{-j\sqrt{2}\beta_o(-x+\sqrt{2}z)/\sqrt{3}}\, (\vec{a}_y) \text{ V/m}$$

Examples

The Optical Fiber.

The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface.



$$\theta_i = \theta_c = 90 - \theta_b$$

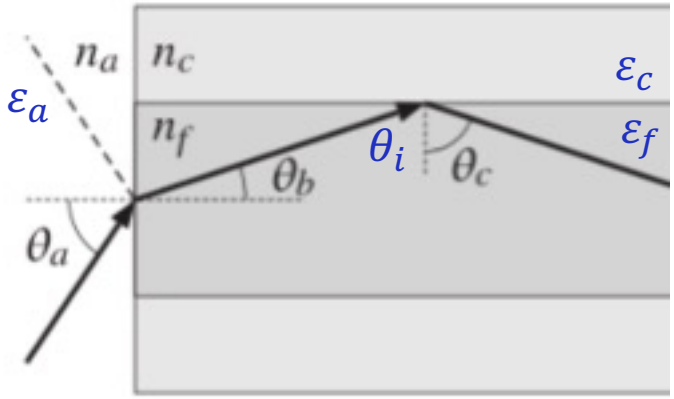
Snell's Law $\sqrt{\varepsilon_{ra}}\sin\theta_a = \sqrt{\varepsilon_{rf}}\sin\theta_b$

$$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin\theta_b = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin(90-\theta_c)$$

$$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\cos\theta_c = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sqrt{1-(\sin\theta_c)^2}$$

Critical angle condition

$$\sin\theta_c = \frac{\sqrt{\varepsilon_{rc}}}{\sqrt{\varepsilon_{rf}}} \Rightarrow \sin\theta_a = \frac{\sqrt{\varepsilon_{rf}-\varepsilon_{rc}}}{\sqrt{\varepsilon_{ra}}}$$



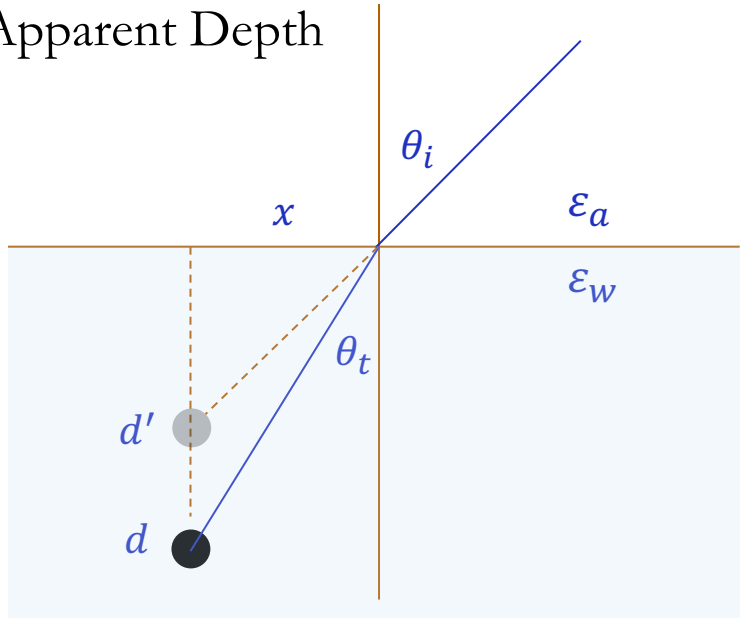
$$\varepsilon_{ra} = 1, \varepsilon_{rf} = 2.22, \varepsilon_{rc} = 2.19$$

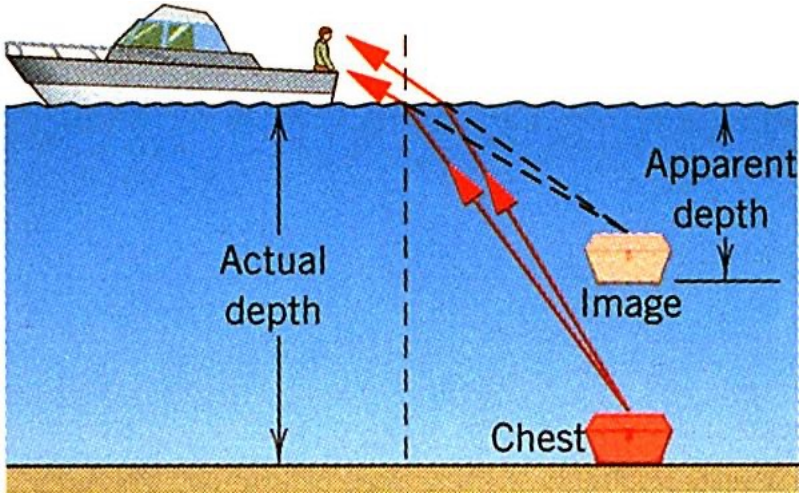
$$\theta_c = 83.4°$$

$$\theta_a = 9.9°$$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

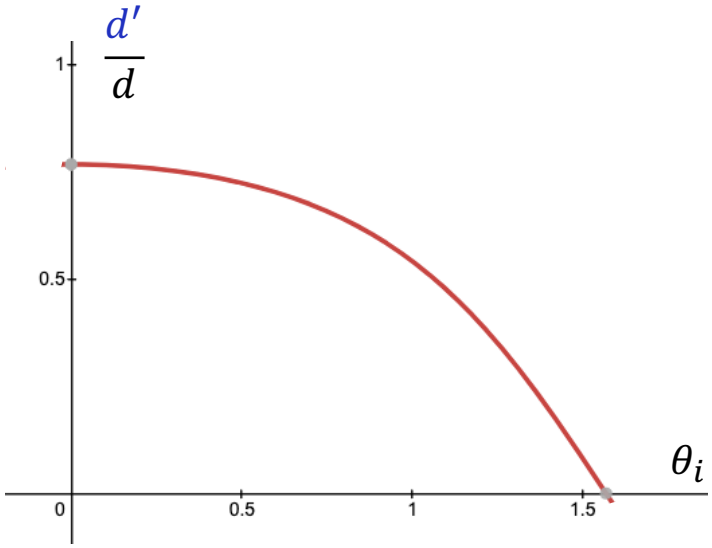
$$d' = x \cot\theta_i$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\sqrt{\varepsilon_{ra}}\sin\theta_i = \sqrt{\varepsilon_{rw}}\sin\theta_t$$

$$\frac{d'}{d} = \frac{\sqrt{\varepsilon_{ra}}}{\sqrt{\varepsilon_{rw}}} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{\varepsilon_{rw} - (\sin\theta_i)^2}}$$



EM Waves Reflection and Transmission

The following subjects will be covered

- Double Interface Reflection/transmission.
- Multi-layers Reflection.

Single Slab (double-interface) Problem

Let's consider normal incidence only. All media are lossless.

The basic equations are $\Gamma = \dfrac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ $\eta = \dfrac{E}{H}$

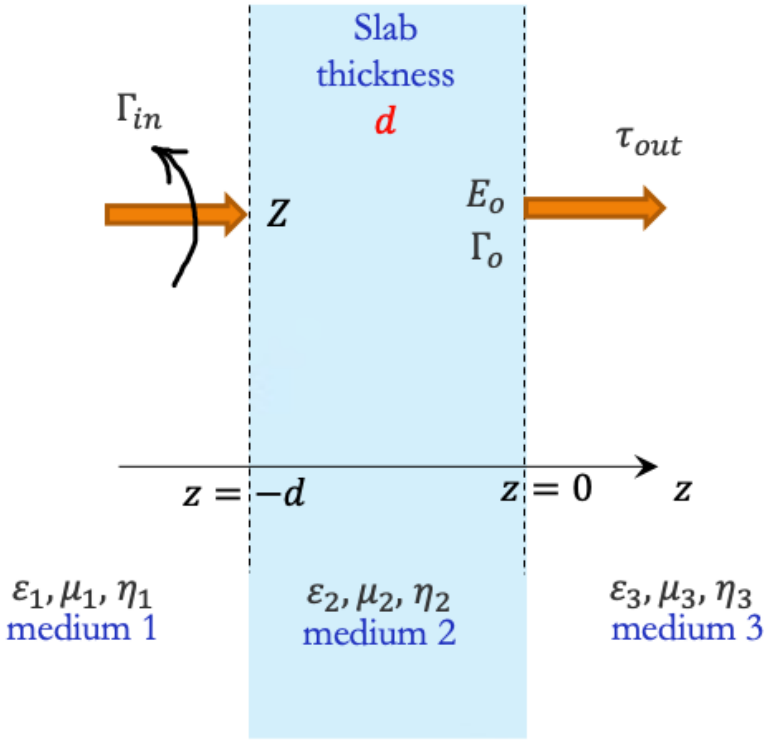
We can write $\Gamma_{in} = \dfrac{Z - \eta_1}{Z + \eta_1}$

The impedance at $z = -d$ is

$$Z = \frac{E^{tot}}{H^{tot}} = \frac{E^i + E^r}{H^i + H^r}$$

$$= \frac{E_o\, e^{-j\beta_2(-d)} + \Gamma_o E_o\, e^{j\beta_2(-d)}}{\frac{E_o}{\eta_2} e^{-j\beta_2(-d)} - \Gamma_o \frac{E_o}{\eta_2} e^{j\beta_2(-d)}}$$

$$= \eta_2 \frac{1 + \Gamma_o\, e^{-j2\beta_2 d}}{1 - \Gamma_o\, e^{-j2\beta_2 d}}$$



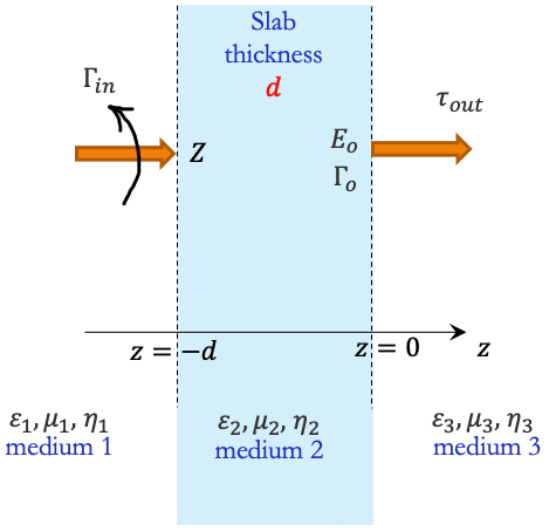
Substitute $\Gamma_o = \dfrac{\eta_3 - \eta_2}{\eta_3 + \eta_2}$ ⇒

$$Z = \eta_2 \left\{ \frac{(\eta_3 + \eta_2) + (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}}{(\eta_3 + \eta_2) - (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}} \right\}$$

Single Slab Problem

The input impedance is

$$Z = \eta_2 \left\{ \frac{(\eta_3 + \eta_2) + (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}}{(\eta_3 + \eta_2) - (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}} \right\}$$



Now can evaluate the overall reflection coefficient.

$$\Gamma_{in} = \frac{Z - \eta_1}{Z + \eta_1}$$

$$= \frac{\eta_2\left[(\eta_3 + \eta_2) + (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}\right] - \eta_1\left[(\eta_3 + \eta_2) - (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}\right]}{\eta_2\left[(\eta_3 + \eta_2) + (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}\right] + \eta_1\left[(\eta_3 + \eta_2) - (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}\right]}$$

$$= \frac{\Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2\beta_2 d}} \qquad \Gamma_{12} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \qquad \Gamma_{23} = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2}$$

$$\approx \Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d} \quad \text{for} \quad \Gamma_{12} \text{ and } \Gamma_{23} \text{ samll}$$

Similarly, $\tau_{out} = \dfrac{\tau_{12}\tau_{23}\, e^{-j\beta_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2\beta_2 d}}$ $\qquad \tau_{12} = 1 + \Gamma_{12}$ $\qquad \tau_{23} = 1 + \Gamma_{23}$

For a lossless slab

Power flux in=Power flux out

$$\frac{1}{\eta_1}(1 - |\Gamma_{in}|^2) = \frac{1}{\eta_3}|\tau_{out}|^2$$

$$(1 - |\Gamma_{in}|^2) = \frac{\eta_1}{\eta_3}\,|\tau_{out}|^2$$

Transmittance

Examples

1) A 10 GHz UPW is normally incident on a dielectric slab ($\varepsilon_r = 2.56$) of thickness d, with air on both sides of the slab. Find the required thickness for zero overall reflection at 10 GHz.

$$\Gamma_{in} = 0 \Rightarrow \left|\Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}\right| = 0 \qquad \Gamma_{12} = -\Gamma_{23} \Rightarrow |\Gamma_{12}|\left|1 - e^{-j2\beta_2 d}\right| = 0$$

$$e^{-j2\beta_2 d} = 1 = \cos(2\beta_2 d) - j\sin(2\beta_2 d) \Rightarrow 2\beta_2 d = n(2\pi), \quad n = 0,1,2,3,\ldots$$

$$d = n\left(\frac{\pi}{\beta_2}\right), \quad 0, \frac{\pi}{\beta_2}, \frac{2\pi}{\beta_2}, \ldots$$

$$d = n\left(\frac{\lambda_2}{2}\right), \quad 0, \frac{\lambda_2}{2}, \lambda_2, \frac{3\lambda_2}{2}, \ldots$$

Multiples of half wavelength

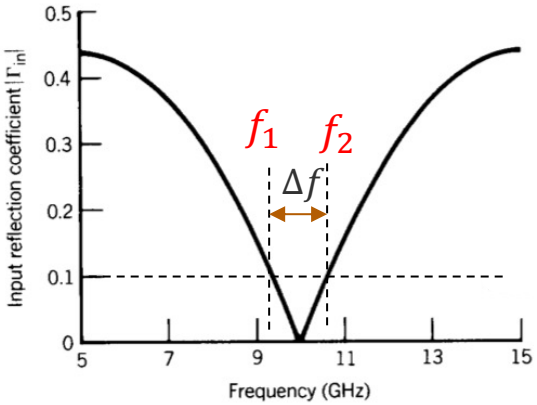
At 10 GHz, $\lambda_2 = \dfrac{c}{\sqrt{\varepsilon_{r2}} \cdot f} = 1.875\ cm \Rightarrow d = 0.9375\ cm$

Half-wavelength Transformer

How much reflection for 5 GHz $\Gamma_{in} = |\Gamma_{12}|\left|1 - e^{-j2\beta_2 d}\right| =$

$$(0.231)\left|1 - e^{-j2\omega\sqrt{2.56}\frac{0.009375}{c}}\right| = 0.438$$

Application: Radome Design



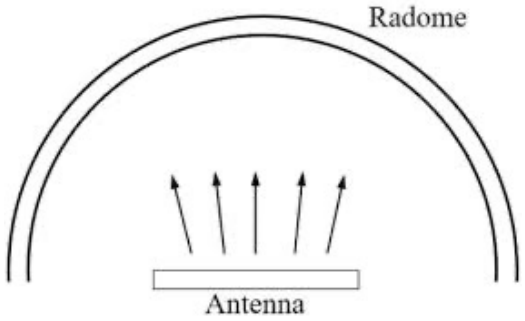
Find the bandwidth for 10% field reflection

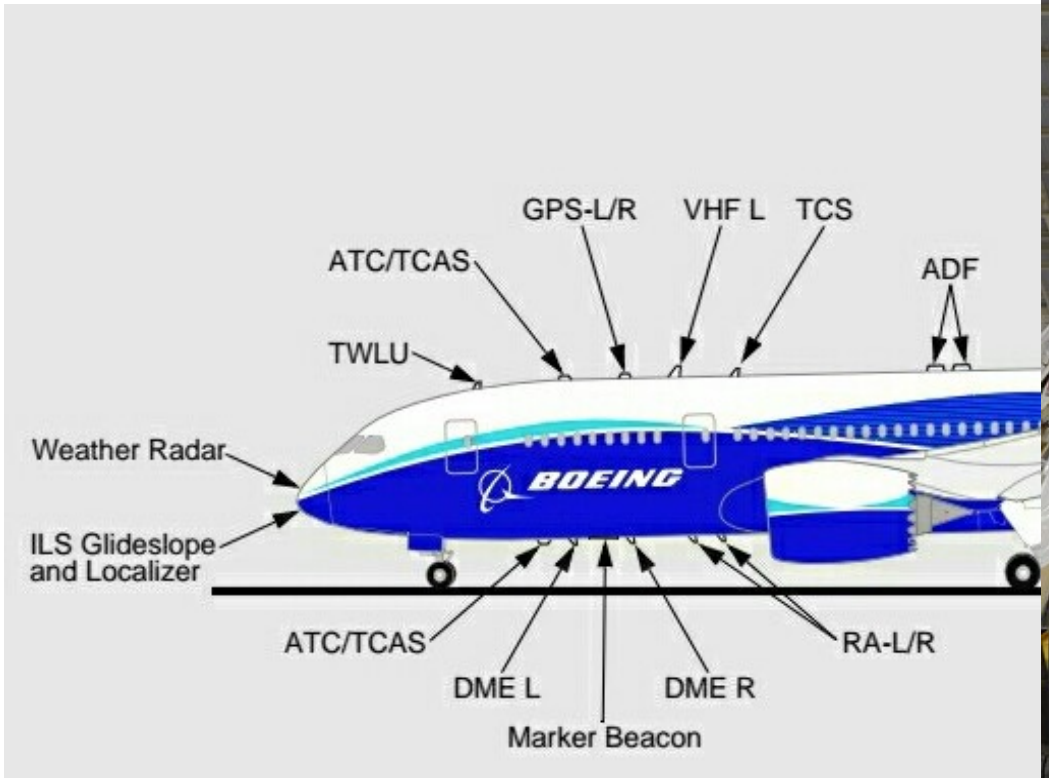
$$(0.231)\left|1 - e^{-j2\beta_2 d}\right| = 0.1$$

$f_1 = 9.3\ GHz,\ f_2 = 10.7\ GHz, BW = \Delta f = 1.4\ GHz$

Factional BW $\Delta f / f = 14\%$

Radome Design







Examples

2) A coating is needed on a building walls ($\varepsilon_{rw} = 4$) such that incident waves at 10 GHz from air are not reflected. Find the required coating thickness and material.

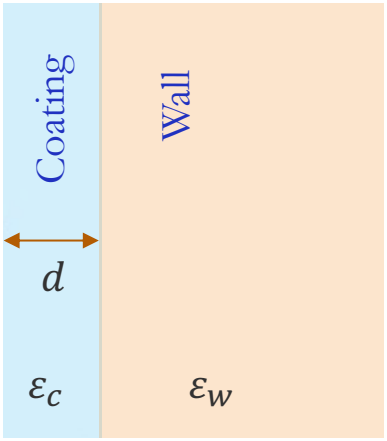
$\Gamma_{in} = 0$ ➡ $\left|\Gamma_{ac} + \Gamma_{cw}\, e^{-j2\beta_c d}\right| = 0$

$$\Gamma_{ac} = \frac{\eta_c - \eta_a}{\eta_c + \eta_a}$$

$$\Gamma_{cw} = \frac{\eta_w - \eta_c}{\eta_w + \eta_c}$$

Look at $2\beta_c d = 2\frac{2\pi}{\lambda_c} d = \pi \frac{4}{\lambda_c} d$

If we take $d = \frac{\lambda_c}{4}$ then $2\beta_c d = \pi$



➡ $e^{-j2\beta_c d} = e^{-j\pi} = -1$ ➡

$$\left|\frac{\eta_c - \eta_a}{\eta_c + \eta_a} - \frac{\eta_w - \eta_c}{\eta_w + \eta_c}\right| = 0$$

$$\left|\frac{(\eta_c - \eta_a)(\eta_w + \eta_c) - (\eta_w - \eta_c)(\eta_c + \eta_a)}{(\eta_c + \eta_a)(\eta_w + \eta_c)}\right| = \left|\frac{2(\eta_c^2 - \eta_a\eta_w)}{(\eta_c + \eta_a)(\eta_w + \eta_c)}\right| = 0 \;\Rightarrow\; 2|(\eta_c^2 - \eta_a\eta_w)| = 0$$

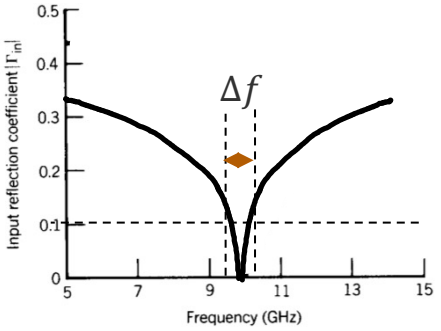
If we take $\eta_c = \sqrt{\eta_a \eta_w}$ ➡ $\Gamma_{in} = 0$

$$d = \frac{\lambda_c}{4} = 0.53\ cm$$

Also $\varepsilon_{rc} = \sqrt{\varepsilon_{ra}\varepsilon_{rw}}$

Quarter-wavelength Transformer

Anti-reflection Coating

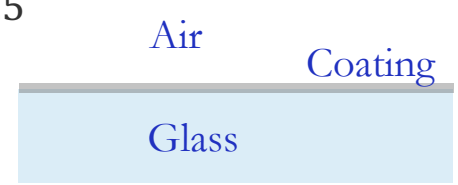


Examples

3) Design a ¼-wavelength antireflection coating for a glass substrate ($\varepsilon_{rg} = 2.25$) . The coating is supposed to work for mid-visible range ($\lambda \approx 550\ nm$) .

Using the quarter wavelength condition $\varepsilon_{rc} = \sqrt{\varepsilon_{ra}\varepsilon_{rg}} = \sqrt{2.25} = 1.5$

$$d = \frac{\lambda_c}{4} = 112.7\ nm$$



The solution is not *that* easy!

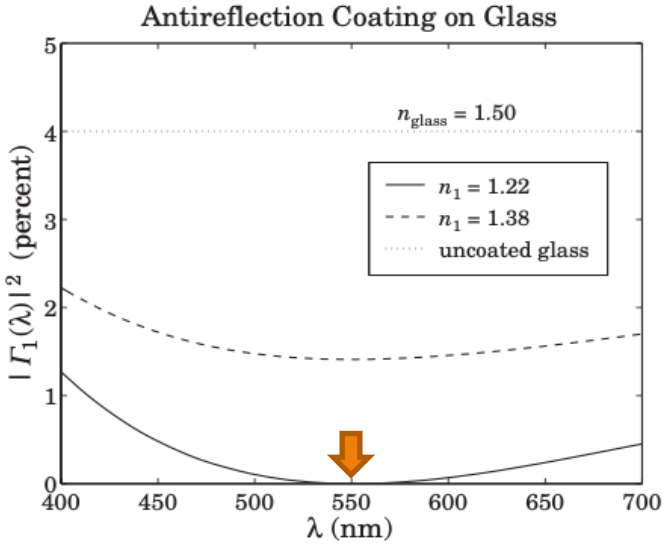
Practically, the closest materials to the requirement are Cryolite ($\varepsilon_r = 1.82$) and Magnesium Fluoride ($\varepsilon_r = 1.91$, more durable for this task and more abundant).

$$\Gamma_{in} = \frac{\sqrt{\varepsilon_{ra}\varepsilon_{rg}} - \varepsilon_{rc}}{\sqrt{\varepsilon_{ra}\varepsilon_{rg}} + \varepsilon_{rc}} = -0.118$$

$$R = |\Gamma_{in}|^2 = 1.4\%$$

Without coating $R = 4\%$

$$d = \frac{\lambda_c}{4} = 99.6\ nm$$



Summary for Zero Reflection

Starting with $\Gamma_{in} = 0 \Rightarrow \left|\Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}\right| = 0$

$$\left| e^{-j2\beta_2 d}\right| = -\frac{\Gamma_{12}}{\Gamma_{23}} \Rightarrow \left| e^{j2\beta_2 d}\right| = -\frac{\Gamma_{23}}{\Gamma_{12}}$$

Case (1) $\Gamma_{12} = \Gamma_{23} \Rightarrow \eta_2^2 = \eta_1\,\eta_3 \Rightarrow e^{j2\beta_2 d} = -1$ Satisfied if $d = (2n+1)\frac{\lambda_2}{4}$

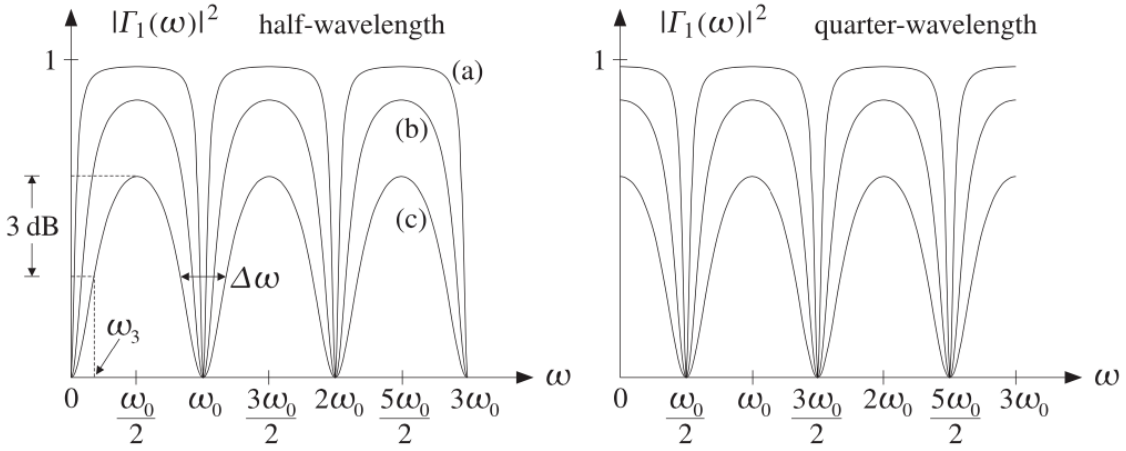
Mediums 1 and 3 arbitrary

¼-wavelength design

Case (2) $\Gamma_{12} = -\Gamma_{23} \Rightarrow \eta_1 = \eta_3 \Rightarrow e^{j2\beta_2 d} = +1$ Satisfied if $d = n\frac{\lambda_2}{2}$

Medium 2 arbitrary

½ -wavelength design



$\Gamma_{12} = 0.9\,(a), 0.7\,(b), 0.5\,(c)$

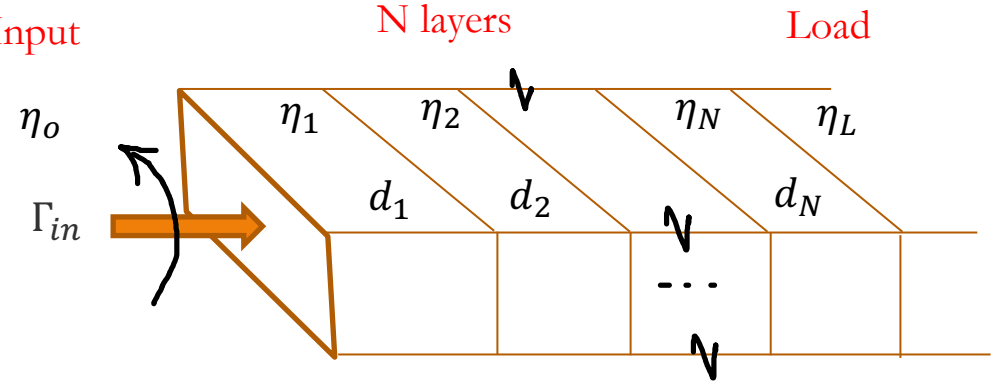
$$|\Gamma_{in}|^2{}_{max} = \frac{4\,\Gamma_{12}{}^2}{(1+\Gamma_{12}{}^2)^2}$$

Multi-layer Reflection

Let's consider normal incidence only. All media are lossless.

We use an extension of the single slab result.

Assume all $|\Gamma| \ll 1$



$$\Gamma_{in} \approx \Gamma_o + \Gamma_1\, e^{-j2\beta_1 d_1} + \Gamma_2\, e^{-j2(\beta_1 d_1 + \beta_2 d_2)} + \cdots + \Gamma_N\, e^{-j2(\beta_1 d_1 + \cdots + \beta_N d_N)}$$

Using a quarter-wavelength design

$$\beta_1 d_1(f) = \frac{2\pi}{\lambda_1}\left(\frac{\lambda_{o1}}{4}\right) = \frac{\pi}{2}\left(\frac{f}{f_o}\right) = \theta \qquad k_n d_n(f) = \theta$$

$$\Gamma_{in}(f) \approx \Gamma_o + \Gamma_1\, e^{-j2\theta} + \Gamma_2\, e^{-j4\theta} + \cdots + \Gamma_N\, e^{-j2N\theta}$$

$$\Gamma_{in}(f) \approx \sum_{n=0}^{N} \Gamma_n\, e^{-j2n\theta}$$

We can fit this function to any of the known flat passband expansions

$$\Gamma_o = \Gamma_{o1} = \frac{\eta_1 - \eta_o}{\eta_1 + \eta_o}$$

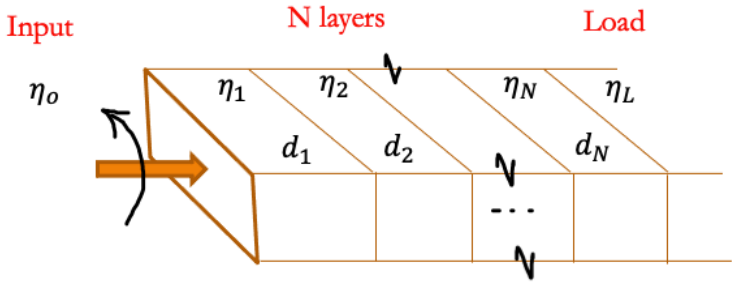
$$\Gamma_n = \Gamma_{n,n+1} = \frac{\eta_{n+1} - \eta_n}{\eta_{n+1} + \eta_n}$$

$$\Gamma_N = \Gamma_{NL} = \frac{\eta_L - \eta_N}{\eta_L + \eta_N}$$

Multi-layer Reflection

For example, the Binomial expansion.

$$B(\theta) = \sum_{n=0}^{N} 2^{-N} \frac{N!}{n!\,(N-n)!}\, e^{-j2n\theta} = e^{-jN\theta} \cos^N(\theta)$$



Take $$\Gamma_n = 2^{-N} \frac{\eta_L - \eta_o}{\eta_L + \eta_o} \frac{N!}{n!\,(N-n)!}$$

$$|\Gamma_{in}|(f) = \frac{\eta_L - \eta_o}{\eta_L + \eta_o} \cos^N\left(\frac{\pi}{2}\frac{f}{f_o}\right)$$

The fractional bandwidth is given by $$\frac{\Delta f}{f_o} = 2 - \frac{4}{\pi} \cos^{-1} \left| \frac{\Gamma_m}{(\eta_L - \eta_o)/(\eta_L + \eta_o)} \right|^{1/N}$$

where Γ_m is the maximum allowed overall reflection (a design parameter).

Multi-layer Reflection

Example: It is needed to match air to a dielectric material of $\varepsilon_r = 4$ using two slabs of quarter wavelength. A fractional bandwidth of no less than 37.5% is needed around 10 GHz. Use Binomial expansion.

$$\Gamma_o = 2^{-2}\frac{\eta_o/2-\eta_o}{\eta_o/2+\eta_o}\frac{2!}{0!\,(2-0)!} = -\frac{1}{12} \quad \Rightarrow \quad \frac{\eta_1-\eta_o}{\eta_1+\eta_o} = -\frac{1}{12} \quad \Rightarrow \quad \eta_1 = \frac{11}{13}\eta_o$$

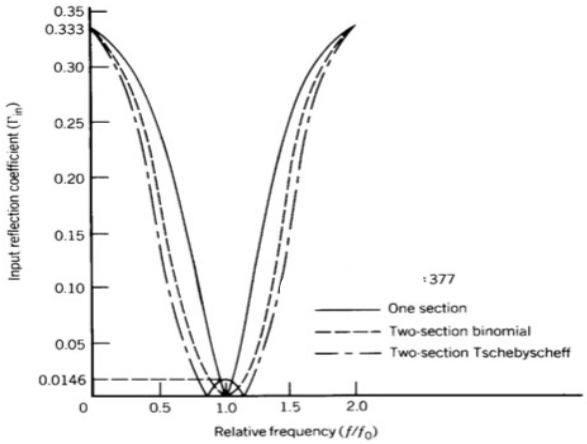
$$\varepsilon_{r1} = (13/11)^2 = 1.4 \qquad d_1 = \frac{\lambda_{o1}}{4} = \frac{\lambda_o}{4\sqrt{1.4}} = 0.635\ cm$$

$$\Gamma_1 = 2^{-2}\frac{\eta_o/2-\eta_o}{\eta_o/2+\eta_o}\frac{2!}{1!\,(2-1)!} = -\frac{1}{6} \quad \Rightarrow \quad \frac{\eta_2-\eta_1}{\eta_2+\eta_1} = -\frac{1}{6} \quad \Rightarrow \quad \eta_2 = \frac{5}{7}\eta_o$$

$$\varepsilon_{r2} = (7/5)^2 = 1.96 \qquad d_2 = \frac{\lambda_{o2}}{4} = \frac{\lambda_o}{4\sqrt{1.96}} = 0.535\ cm$$

$$\frac{\Delta f}{f_o} = 0.375 = 2 - \frac{4}{\pi}\cos^{-1}\left|\frac{\Gamma_m}{\frac{\eta_o/2-\eta_o}{\eta_o/2+\eta_o}}\right|^{1/2} \quad \Rightarrow \quad \Gamma_m = 0.028$$

$$|\Gamma_{in}|(f) = \frac{1}{3}\cos^2\left(\frac{\pi}{2}\frac{f}{f_o}\right)$$



Q & A

Send me an email if you have any questions.